EE499 FYP — Simulation Findings Report

Output Capacitor Revision & BOM Alignment | Phase 2 Simulation Campaign

Field	Detail
Author	Fortune Olose — BEng Electronic Engineering, Maynooth University
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Module	EE499 Final Year Project
Date	1 April 2026
Revision	Simulation File: LTC1871_BOM_Caps_Rev4.asc | Log date: Wed 1 Apr 2026 15:55
IC / Topology	LTC1871 (MSOP-10) — Boost Converter, CCM, 350 kHz
Deadline	8 May 2026

1. Executive Summary

This report documents the simulation activity conducted on 1 April 2026 as part of the EE499 FYP Phase 2 campaign. The session had two primary objectives: (1) align the LTspice output capacitor model with the actual Bill of Materials ceramic capacitor bank, and (2) correctly implement Dr. Lawlor's revised simulation configuration (dual-pulse PWL load profile and extended 10 ms transient window). Four simulation revisions were executed within the session. The final revision — LTC1871_BOM_Caps_Rev4.asc — produced clean measurements across all ten metrics. All design specifications are met with margin. The ripple specification failure observed in the preliminary BOM run was resolved by adding two additional 22 µF ceramic capacitors to the output bank.

2. Background and Session Objectives

2.1 Context

The Phase 2 LTspice simulation campaign was substantially completed prior to this session, with all eight headline metrics passing their specifications using a 2×150 µF polycapacitor output bank (LTC1871_Improved_Model.asc). Dr. Lawlor reviewed the submitted work and provided two key modifications to the simulation file (LTC1871_Improved_Model_BL.asc):

- **PWL load profile revision:** two discrete motor-inrush pulses (1.5 A at t = 5.5 ms, 2 A at t = 8.5 ms) replacing the single permanent step, more accurately modelling a shared robotic battery bus.
- **Extended transient window:** .tran 0 10m 5m, storing only the post-startup 5–10 ms range to ensure all measurements are taken in fully settled steady state.

A separate BOM cap model (LTC1871_BOM_Caps.asc) had been created to replace the 2×150 µF polycap placeholder with the actual hardware ceramic bank (4×22 µF + 2×10 µF). Running this model revealed a ripple specification failure: 275.8 mV (2.26%) against the 240 mV (2%) design target, caused by the lower effective capacitance of the ceramic bank (~74 µF derated) relative to the polycap placeholder (~300 µF).

The objectives for this session were therefore:

- **Fix the ripple failure** by increasing the ceramic output capacitor count.
- **Implement Dr. Lawlor's .asc modifications** (dual-pulse PWL, 10 ms tran window) in the BOM cap model.

- **Correct the .meas time windows** to reference the relative time axis used by LTspice when Tstart > 0.
- **Produce a final verified result** across all ten metrics.

3. Simulation Revision Log

Four simulation revisions were executed during this session. Each is documented below with the change rationale and outcome.

3.1 LTC1871_BOM_Caps.asc (Preliminary — Pre-session)

Parameter	Detail
Cap bank	4×22 µF GRM31CR61E226M + 2×10 µF GRM21BR61E106K (nominal 108 µF, ~74 µF derated)
PWL profile	Single step: 0.5 A → 2.0 A at t = 3 ms (old profile)
Tran window	.tran 5m startup (old — 0–5 ms)
.meas windows	FROM 2m TO 5m (old windows, matching old tran)
Outcome	RIPPLE FAIL: 275.8 mV (2.26%) vs 240 mV target. All other metrics pass.

3.2 LTC1871_BOM_Caps_Rev3.asc

Parameter	Detail
Change 1	Cap bank increased: 6×22 µF C1a (132 µF nominal, Rser=0.21 mΩ) + 2×10 µF C1b (20 µF nominal, Rser=1.25 mΩ). Total: 152 µF nominal, ~104 µF derated.
Change 2	.tran updated to 0 10m 5m per Dr. Lawlor.
Change 3	Dual-pulse PWL load profile implemented per Dr. Lawlor.
Change 4	.meas windows updated — error: referenced absolute times (e.g. FROM 5m) not relative times.
Outcome	All 10 measurements FAILED. Circuit solved correctly (DC_dev returned 0.0659) but .meas FROM/TO values fell outside the saved time range. Root cause: LTspice stores time relative to Tstart (5 ms), so all .meas references must be offset by −5 ms.

3.3 LTC1871_BOM_Caps_Rev4.asc (Final — Verified)

Parameter	Detail
Change	All .meas FROM/TO and TD= values corrected by subtracting Tstart (5 ms). E.g., Vout_droop2 FROM 3.5m (absolute: 8.5 ms) TO 4.4m (absolute: 9.4 ms).
Cap bank	Unchanged from Rev3: C1a = 132 µF / 0.21 mΩ, C1b = 20 µF / 1.25 mΩ.
Outcome	All 10 measurements returned clean values. All specifications met.

4. Measurement Window Reference

The table below documents all .meas directives in the final Rev4 file, showing both the relative time (as written in the .asc) and the corresponding absolute simulation time. This is required because .tran 0 10m 5m causes LTspice to store data starting at Tstart = 5 ms, creating a −5 ms offset between file values and wall-clock simulation time.

Measurement	Relative Time (asc)	Absolute Time	Purpose
Vout_before	0 – 0.4 ms	5.0 – 5.4 ms	Pre-pulse steady-state average voltage
Vout_droop1	0.5 – 1.4 ms	5.5 – 6.4 ms	Minimum voltage during 1.5 A motor-inrush pulse
Vout_recov1	TD = 0.5 ms	TD abs = 5.5 ms	Time for Vout to recover to 12.15 V after 1.5 A step
Vout_idle	2.1 – 3.4 ms	7.1 – 8.4 ms	Average voltage in idle window between pulses
Vout_droop2	3.5 – 4.4 ms	8.5 – 9.4 ms	Minimum voltage during 2 A motor-inrush pulse (worst case)
Vout_recov2	TD = 3.5 ms	TD abs = 8.5 ms	Time for Vout to recover to 12.15 V after 2 A step
Vout_ripple	3.6 – 4.4 ms	8.6 – 9.4 ms	Peak-to-peak ripple during 2 A load (worst-case ripple)
IL_peak	0 – 0.4 ms	5.0 – 5.4 ms	Maximum inductor current in pre-pulse window
Pin	3.6 – 4.4 ms	8.6 – 9.4 ms	Average input power during 2 A load
Pout	3.6 – 4.4 ms	8.6 – 9.4 ms	Average output power during 2 A load
DC_dev	0 – 5 ms (rel)	5.0 – 10.0 ms	RMS deviation of Vout from 12.1985 V reference

5. Output Capacitor Bank — Revision

5.1 Problem Statement

The preliminary BOM cap simulation used 4×22 µF + 2×10 µF ceramics, yielding a nominal capacitance of 108 µF and an estimated effective capacitance of ~74 µF after DC-bias derating at 12 V (GRM31CR61E226M: approximately 55% retention at 12 V, 25 V-rated part; GRM21BR61E106K: approximately 65% retention). The resulting peak-to-peak ripple of 275.8 mV exceeded the 240 mV (2%) design specification. Applying the inverse-proportionality of ripple to capacitance:

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C_required = C_existing × (ΔV_measured / ΔV_target) = 74 µF × (275.8 / 240) = 85.1 µF effective
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Adding two further 22 µF capacitors (6×22 µF total) increases nominal capacitance to 132 µF and effective derated capacitance to approximately 104 µF — 22% above the 85.1 µF minimum. Combined with the unchanged 2×10 µF bank, total effective capacitance is approximately 104 µF, yielding a predicted ripple of:

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ΔV_predicted = ΔV_measured × (C_old / C_new) = 275.8 mV × (74 / 104) = 196 mV
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5.2 Revised BOM — Output Capacitor Bank

Ref.	Part Number	Description	Qty	Nom. Cap	Rser (bank)	Est. Derated
C1a	GRM31CR61E226M (Murata)	22 µF 25V 1206 MLCC X5R	6	132 µF	0.21 mΩ	~84 µF @ 12V
C1b	GRM21BR61E106K (Murata)	10 µF 25V 0805 MLCC X5R	2	20 µF	1.25 mΩ	~20 µF @ 12V
Total	—	—	**8**	**152 µF**	—	**~104 µF eff.**

Note: Derating estimates based on Murata SimSurfing data for X5R dielectric at 12 V applied / 25 V rated. Actual derating should be confirmed from the SimSurfing tool before PCB submission.

6. Rev4 Simulation Results — Full Metric Table

All measurements below are taken from the LTC1871_BOM_Caps_Rev4.net log (Wed 1 Apr 2026, 15:55). The simulation ran to completion in 11.72 seconds using the alternate solver with trapezoidal integration. Source stepping was required to find the operating point; Gmin stepping failed as in all prior runs with this model — this is a known, non-pathological behaviour of the LTC1871.sub SPICE model.

Metric	Measured Value	Design Target	Status	Margin
Output Voltage (pre-pulse)	**12.170 V**	12.0 V regulated	**PASS**	*+1.4%*
	Slightly above 12.19 V theory due to measurement window falling in early startup tail. Vout_idle (fully settled) = 12.191 V, within 0.01 V of theory.			
Output Voltage (idle)	**12.191 V**	12.0 V regulated	**PASS**	*+1.6%*
Output Ripple (2A load)	**197.6 mV (1.63%)**	< 240 mV (2%)	**PASS**	*18%*
	Predicted 196 mV; actual 197.6 mV — 0.8% above prediction, confirming model accuracy. Previous BOM cap result (275.8 mV) failed; resolved by adding 2× additional 22 µF capacitors.			
Droop — 1.5A pulse	**73 mV (0.60%)**	< 600 mV (5%)	**PASS**	*8×*
	First motor-inrush event. Recovery in 31.7 µs (relative: 0.032 ms after pulse start).			
Droop — 2A pulse	**102 mV (0.84%)**	< 600 mV (5%)	**PASS**	*6×*
	Worst-case inrush event. Droop 39% larger than 1.5A event, proportional to load magnitude.			
Recovery — 1.5A pulse	**31.7 µs**	< 1 ms	**PASS**	*32×*
Recovery — 2A pulse	**40.6 µs**	< 1 ms	**PASS**	*25×*
	Recovery marginally slower than 1.5A event — current-mode loop bandwidth is consistent across load levels.			
Peak Inductor Current	**3.554 A**	< 3.59 A (XAL6060 limit)	**PASS**	*36 mA*

Metric	Measured Value	Design Target	Status	Margin
	Measured in pre-pulse window at VIN = 9.6V. Worst-case (VIN = 8V) would approach 3.59A analytical limit — still well within 11.2A saturation rating.			
Simulated Efficiency (2A)	**96.0%**	> 85%	**PASS**	*+11 pp*
	Improved from 94.83% in polycap model. Ceramic ESR (0.21 mΩ parallel bank) vs polycap (50 mΩ) reduces capacitor conduction losses by ~1.2 pp. Hardware estimate: 88–92% (PCB parasitics).			
DC RMS Deviation	**66.1 mV RMS**	Informational	**INFO**	—
	RMS(Vout − 12.1985V) = 66.1 mV. Dominated by switching ripple envelope. Consistent with PP ripple of 197.6 mV (RMS ≈ PP/2√2 for triangular ripple ≈ 70 mV).			

7. Key Findings and Engineering Implications

7.1 Capacitor Bank Revision — Engineering Justification

The transition from 4×22 µF + 2×10 µF to 6×22 µF + 2×10 µF is a targeted, minimum-change fix. Adding exactly two additional 22 µF capacitors (the same GRM31CR61E226M part already in the BOM) brings the design back within specification without introducing new component families. The revised bank remains compact (six 1206 + two 0805 MLCCs), maintains a consistent footprint family, and carries only minor cost impact (~€0.90 additional at unit pricing).

The design rationale should be documented in the report's Section 4.6 (Output Capacitor Design) as follows: the initial 4+2 bank was sized analytically for 47.6 µF minimum, providing 1.56× margin at nominal. DC-bias derating reduces effective capacitance to ~74 µF, reducing the margin to 1.56× — insufficient once LTspice simulation revealed the actual ripple. The revised 6+2 bank provides ~104 µF effective, restoring the margin to 2.18× and bringing simulated ripple to 197.6 mV with 18% headroom against the 240 mV specification.

7.2 Efficiency Improvement from Ceramic vs Polycap

The simulated efficiency increased from 94.83% (polycap placeholder, Rser = 50 mΩ) to 96.0% (ceramic bank, Rser = 0.21 mΩ parallel bank). The efficiency gain of approximately 1.2 percentage points is attributable to the dramatically lower ESR of the ceramic bank. Capacitor conduction loss scales as I^2_RMS × Rser; the 238× reduction in Rser from polycap to ceramic bank (50 mΩ → 0.21 mΩ) eliminates what was a non-trivial loss contributor in the placeholder model. This finding validates the ceramic cap selection on both ripple and efficiency grounds.

7.3 Dual-Pulse Load Profile — Transient Comparison

Dr. Lawlor's revised PWL profile exposes the converter to two distinct load events: a 1.5 A pulse followed by a 2 A pulse. The results confirm expected behaviour — droop scales with load magnitude (73 mV at 1.5 A vs 102 mV at 2 A), and recovery time is only marginally different (31.7 µs vs 40.6 µs). This consistency demonstrates that the current-mode control loop bandwidth is sufficient across the full load range, and that neither event pushes the converter into an unstable or degraded operating regime.

7.4 Outstanding Simulation Work

The following items remain outstanding and should be prioritised before the May 8 submission deadline:

- **AC loop gain sweep (.ac analysis):** Phase margin and gain margin have not been verified in simulation. This is the single most critical remaining simulation task. The RHPZ at 39.2 kHz constrains crossover to below ~7.8 kHz; the compensation (RC = 8.66 kΩ / CC = 6.8 nF / CC2 = 100 pF) has been analytically justified but not swept.
- **Worst-case line regulation (VIN = 8V):** All results in this report are at VIN = 9.6 V (nominal). The hardest operating point (VIN = 8V, D = 0.333, IL_avg = 3A) must be simulated with the Rev4 cap bank to confirm ripple and droop remain within spec.
- **Startup overshoot investigation:** The 20.8% startup overshoot documented in the Phase 2 report represents a potential hardware kill condition (~14.7V peak). This must be characterised before PCB fabrication.
- **Parametric line regulation sweep:** .step VIN across 8V, 9.6V, 11V to formally report line regulation figure with the BOM cap bank.

8. Simulation File Change Log

File	Date	Status	Key Changes
LTC1871_Improved_Model.asc	Pre-session	Superseded	Original Phase 2 file. 2×150µF polycap. Single step PWL (0.5→2A at 3ms). .tran 5m startup.
LTC1871_Improved_Model_BL.asc	Pre-session	Dr. Lawlor	Supervisor revision. Dual-pulse PWL. .tran 0 10m 5m. DC_dev .meas added. .asc comments added.
LTC1871_BOM_Caps.asc	1 Apr 2026	Superseded	BOM cap bank introduced (4×22µF + 2×10µF). Old measurement windows retained. Ripple FAIL (275.8 mV).
LTC1871_BOM_Caps_Rev3.asc	1 Apr 2026	Superseded	Cap bank increased to 6×22µF + 2×10µF. Dr. Lawlor PWL and tran applied. .meas windows at wrong absolute times — all 10 FAILed.
LTC1871_BOM_Caps_Rev4.asc	1 Apr 2026	**CURRENT**	.meas windows corrected to relative time (−5ms offset from Tstart). All 10 measurements pass. This is the active simulation baseline.

9. SPICE Solver Behaviour Note

All runs in this session required source stepping to find the DC operating point; Gmin stepping failed in every case. This is a known characteristic of the LTC1871.sub model and is not indicative of a circuit problem. The source stepping succeeded in all runs, reaching 100% in approximately 12 seconds. The alternate solver with trapezoidal integration (method=trap) was used throughout, consistent with prior Phase 2 runs. This solver note should be included in the report's simulation methodology section to pre-empt examiner questions about the convergence warnings in the log.

10. Report Updates Required Following This Session

The following sections of the EE499 FYP Report (EE499_FYP_Report_Final.docx) require updating to reflect the findings in this report:

Report Section	Required Update
Section 4.6 — Output Capacitor Design	Update BOM from 4×22µF + 2×10µF to 6×22µF + 2×10µF. Add DC-bias derating analysis justifying the additional capacitors. State new effective capacitance of ~104µF and predicted ripple of ~196mV.
Section 4.9 — Bill of Materials	Update C1 row: change quantity from 4 to 6 for GRM31CR61E226M. Update subtotal. Revise total BOM cost (~€0.90 increase).
Section 6.2 — Line Regulation Results	Add note that results are pending BOM cap re-run at VIN = 8V, 9.6V, 11V parametric sweep.
Section 6.3 — Load Transient Response	Replace single-step results with dual-pulse results (droop1=73mV@1.5A, droop2=102mV@2A). Update recovery times. Note simulation uses Dr. Lawlor's revised PWL profile.
Section 6.5 — Efficiency Analysis	Update efficiency from 94.83% to 96.0%. Add commentary on ceramic vs polycap ESR contribution to efficiency improvement.
Section 6.6 — Results Summary Table	Update ripple row (197.6mV, 1.63%), droop rows (dual values), efficiency row (96.0%). Add footnote referencing this report.
Section 7.4 / Future Work	Add AC loop gain sweep, worst-case VIN=8V simulation, and soft-start overshoot investigation as outstanding items.
Figure 4.1 — LTspice Schematic	Replace with Rev4 schematic screenshot showing C1a and C1b cap bank with correct values and annotation.

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